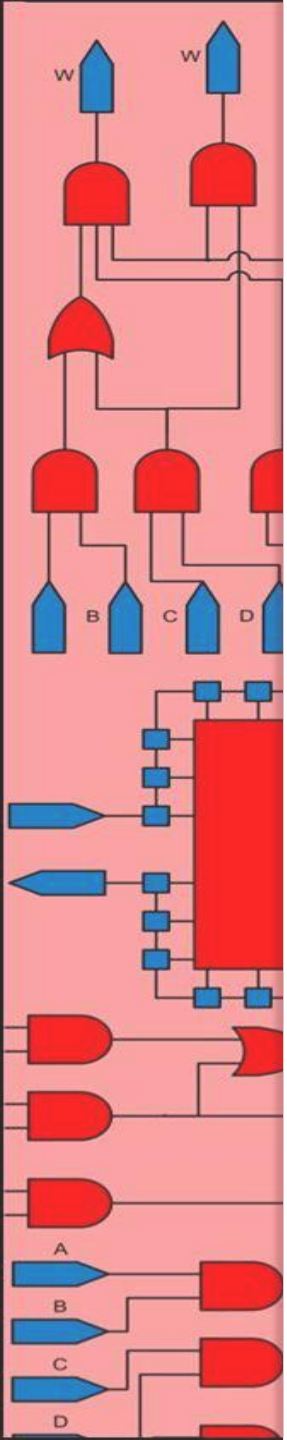


Topic 1

Introduction Switches to Systems



Outline

I- Introduction

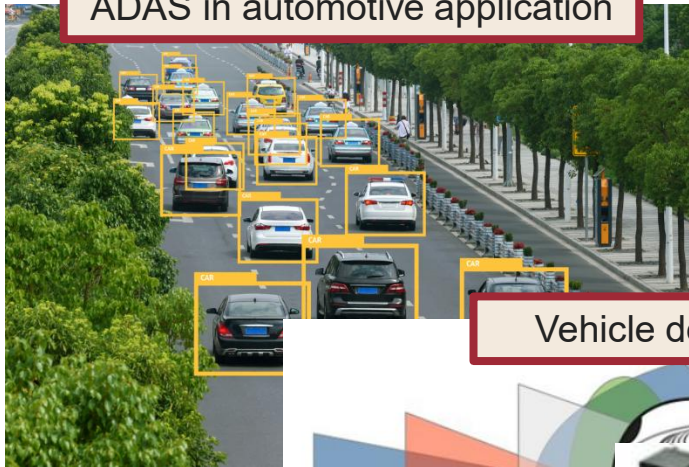
I-1 Hierarchical System Structure

I-2 System description in SystemVerilog

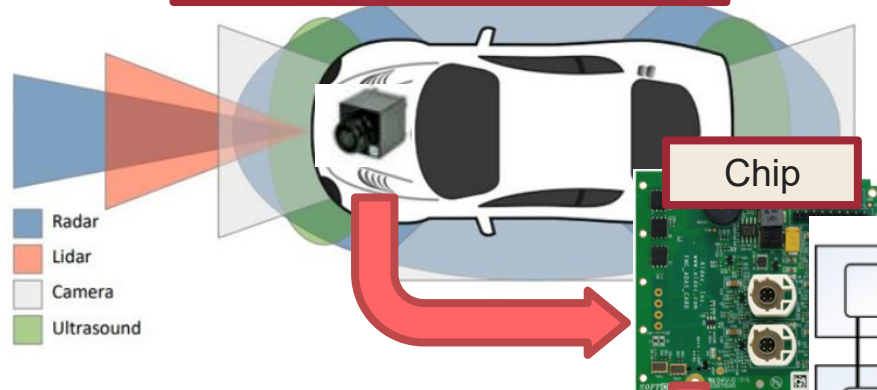
I-3 Testbenches in SystemVerilog

Hierarchical System Structure

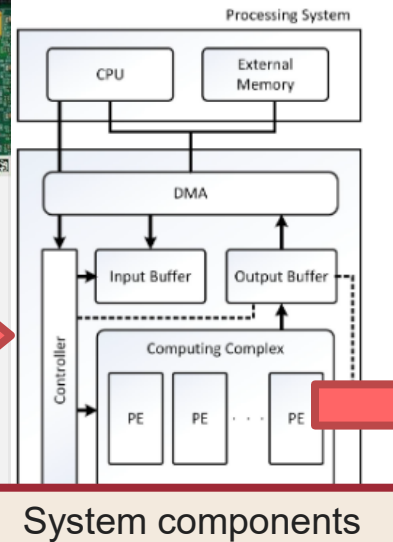
ADAS in automotive application



Vehicle detection in ADAS

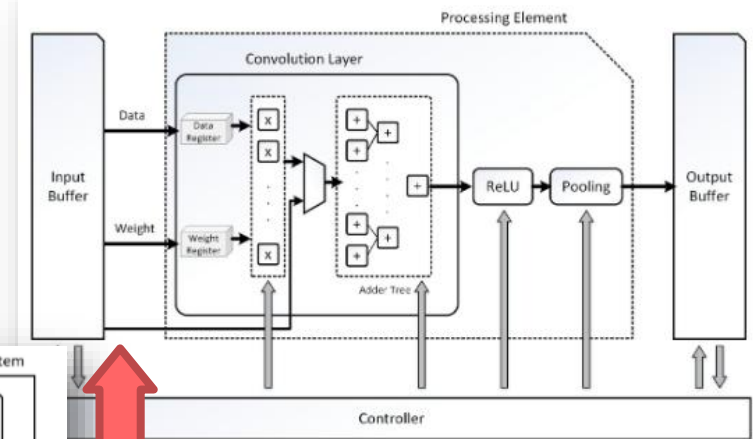


Chip

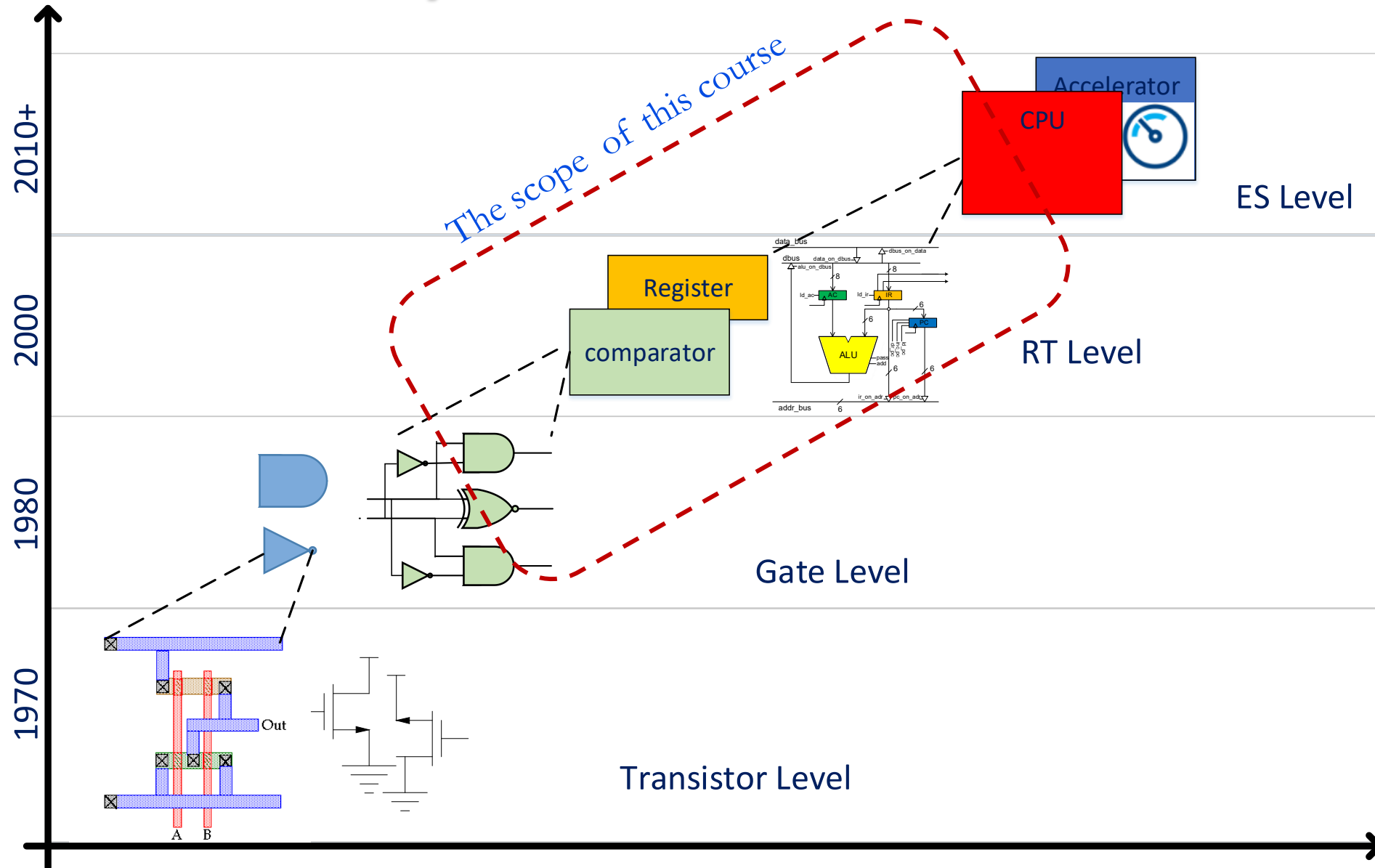


System components

RTL components in Neural network accelerator

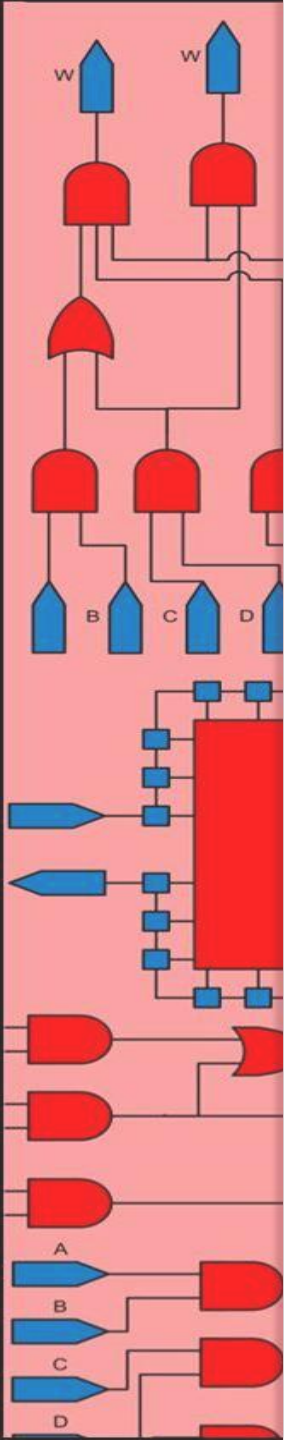


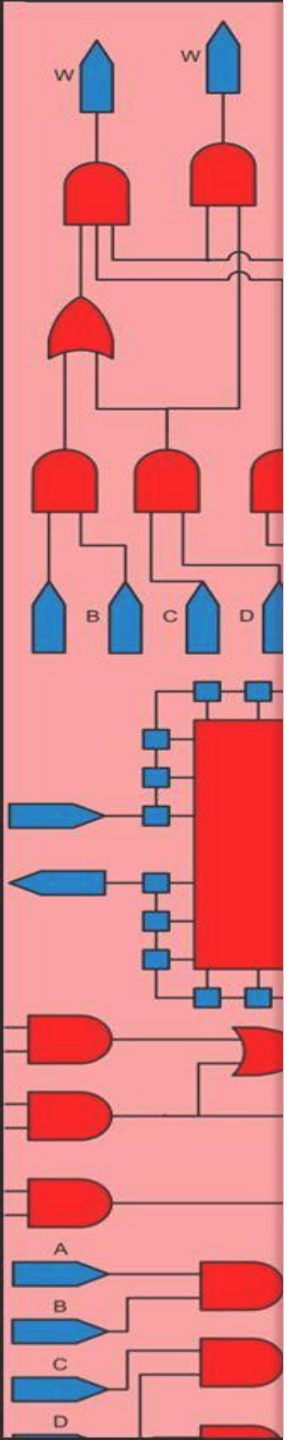
Hierarchical System Structure



Hierarchical System Structure

- System
- RTL subsystem
- RTL components
 - ALU
 - Register file
 - Counter
 - State machines
- Gate Level
- Layout





Hierarchical System Structure

- This is a journey to go from System-level down to transistor-level
- Who will help us through this journey?
- How should we communicate?

System Description in SystemVerilog

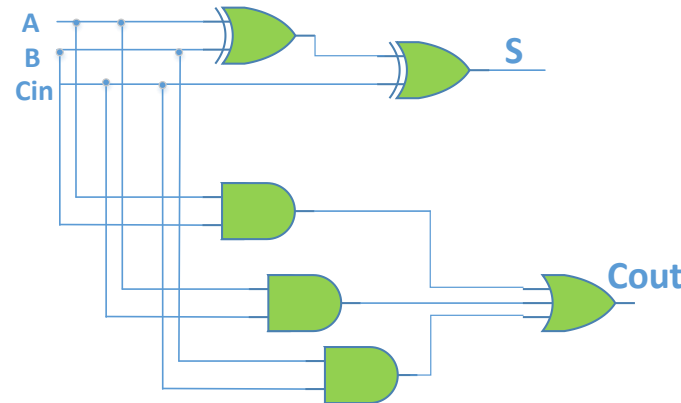
- We need a language to describe the system
- The language should support
I-Behavioral description

```
module full_adder_behavioral ( input  A, B, cin, output Cout, S);  
  
    assign {Cout, S}= A + B+ Cin;  
  
endmodule
```

System Description in SystemVerilog

- We need a language to describe the system
- The language should support
 - 2- Structural description

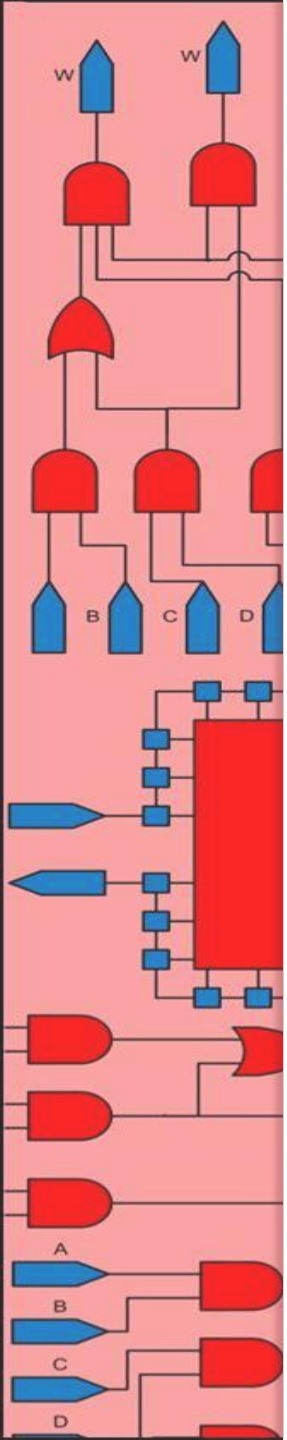
```
module full_adder_structural ( input  A, B, cin, output Cout, S);  
  
    logic W1, W2, W3, W4;  
    xor (W1,A,B);  
    xor (S,W1, Cin);  
  
    and (W2,A,B);  
    and (W3,A,Cin);  
    and (W4,B,Cin);  
    or  (Cout,W2,W3,W4);  
endmodule
```



System Description in SystemVerilog

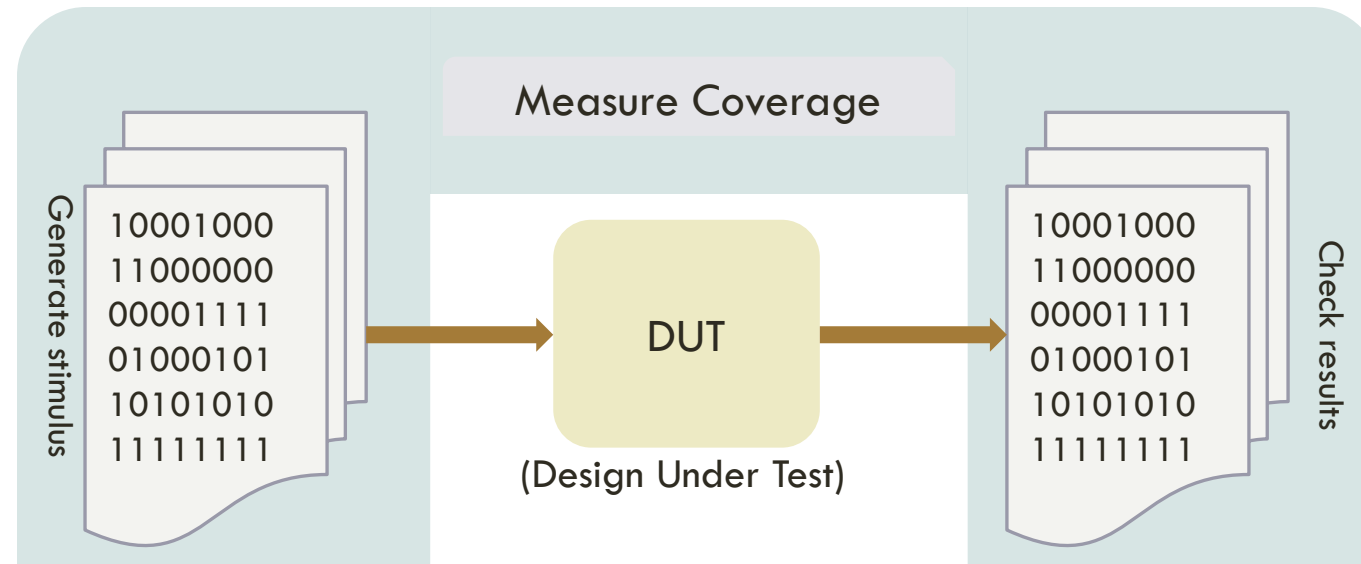
- We need a language to describe the system
- The language should support

3-Can model RTL, gate and transistor levels

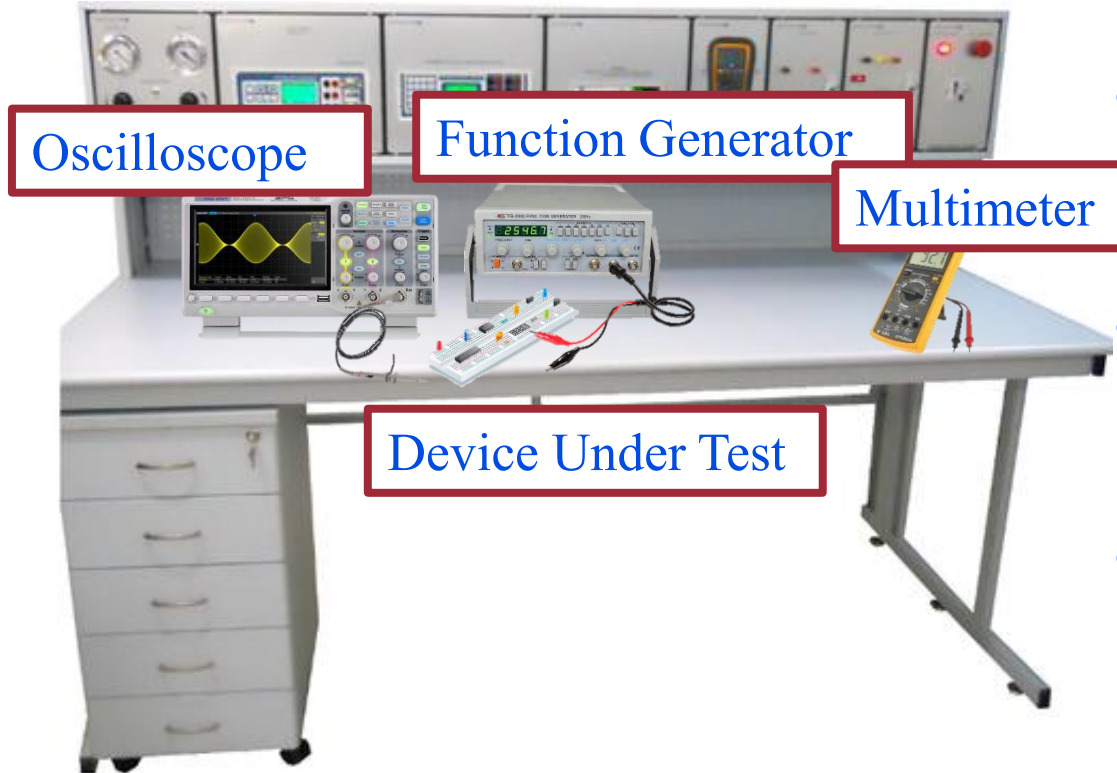


Verification

- Functional test of the correctness of a design with respect to the design specifications.
- Design Verification ask: Did you design the product right?

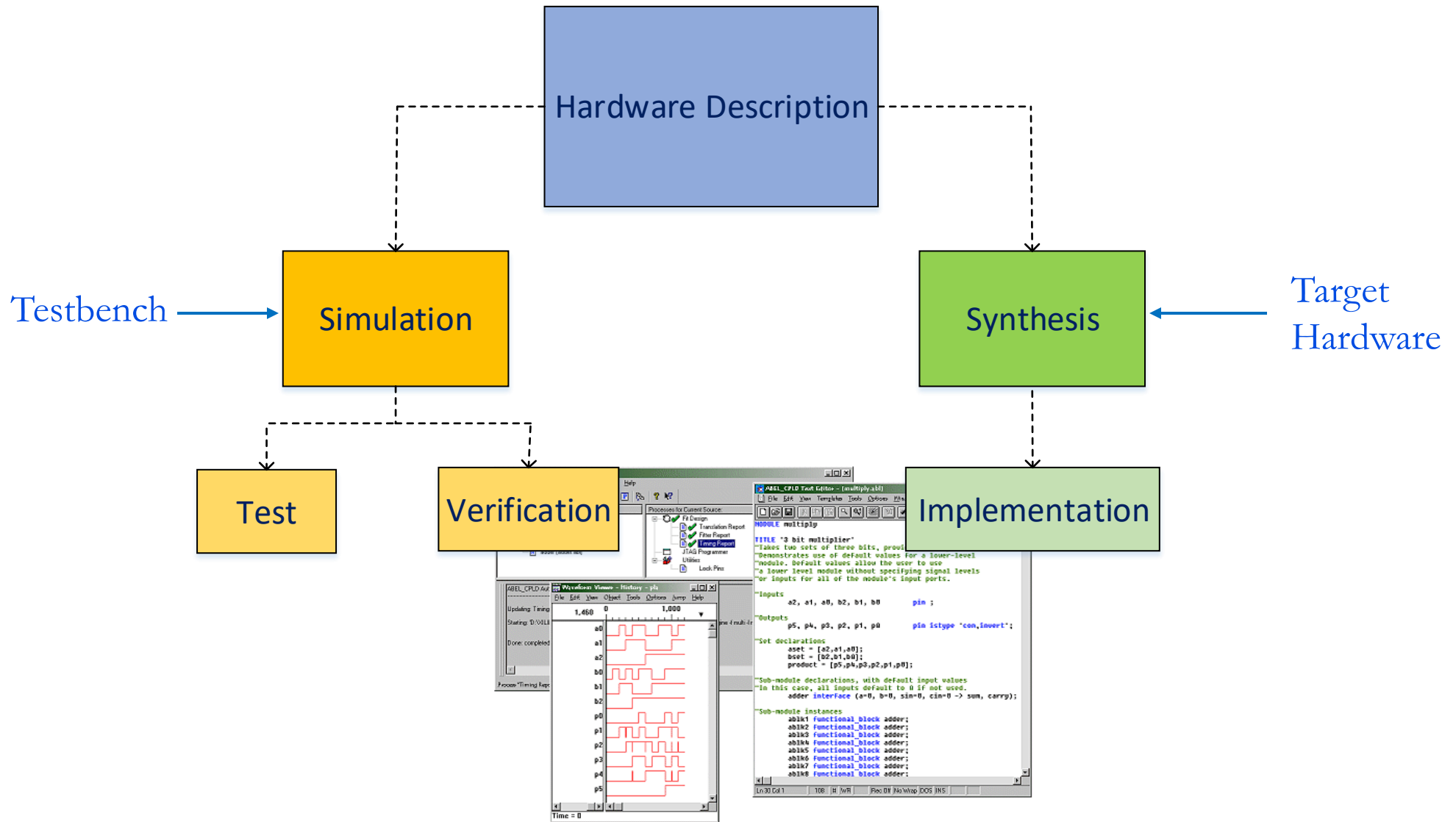


Testbenches in SystemVerilog



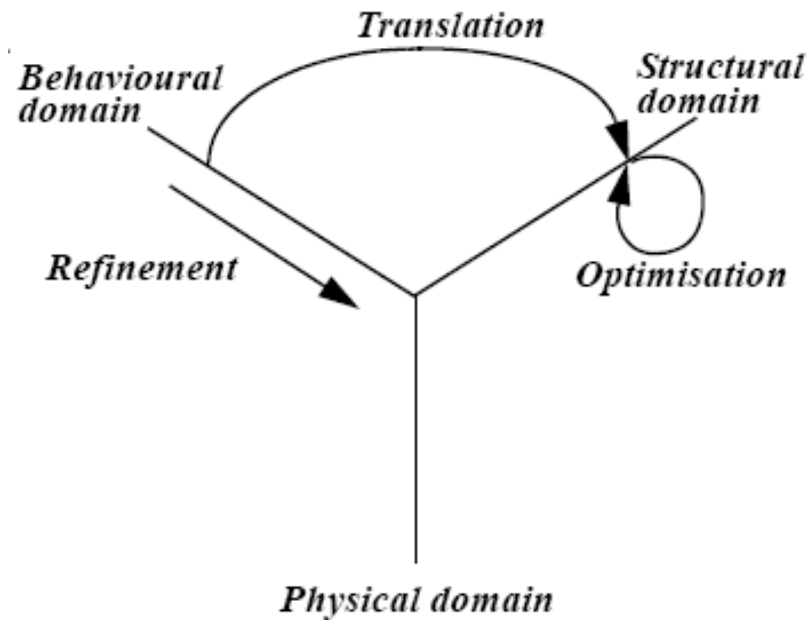
- A testbench contains equipment for input generation
- It includes equipment for response observation and analysis
- Usually output is a waveform

System Description in SystemVerilog



Synthesis

Synthesis is essentially refining a more abstract specification to a relatively more detailed specification

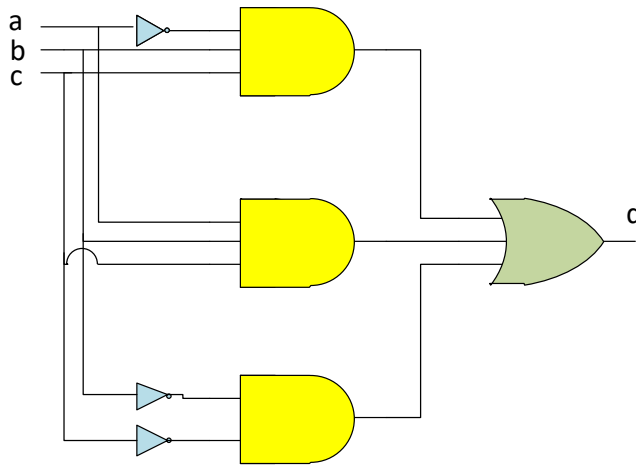


Synthesis
=
Domain Translation
+
Refinement:
Higher Abstraction to Lower Abstraction
+
Optimisation

Synthesis Example

input

```
1 module simple (a,b,c,d);
2   input a,b,c;
3   output d;
4   assign d = (~a & b & c) | (a & b & c) | (~b & ~c);
5 endmodule
```

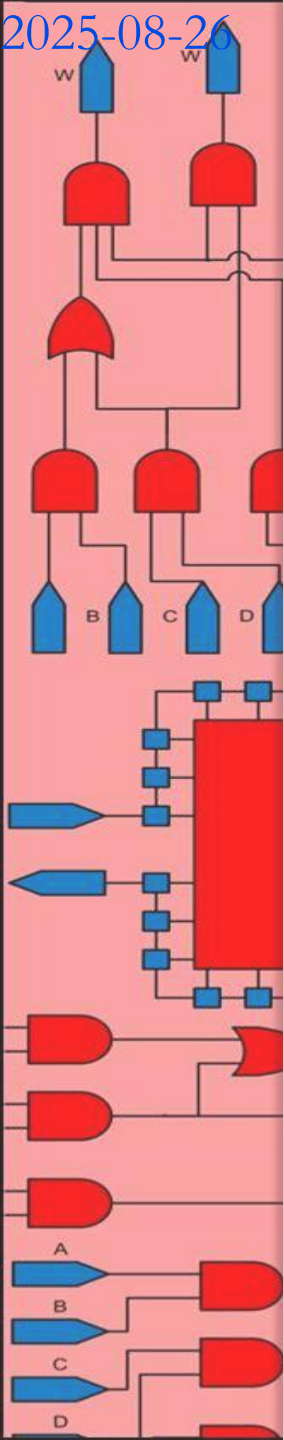


output

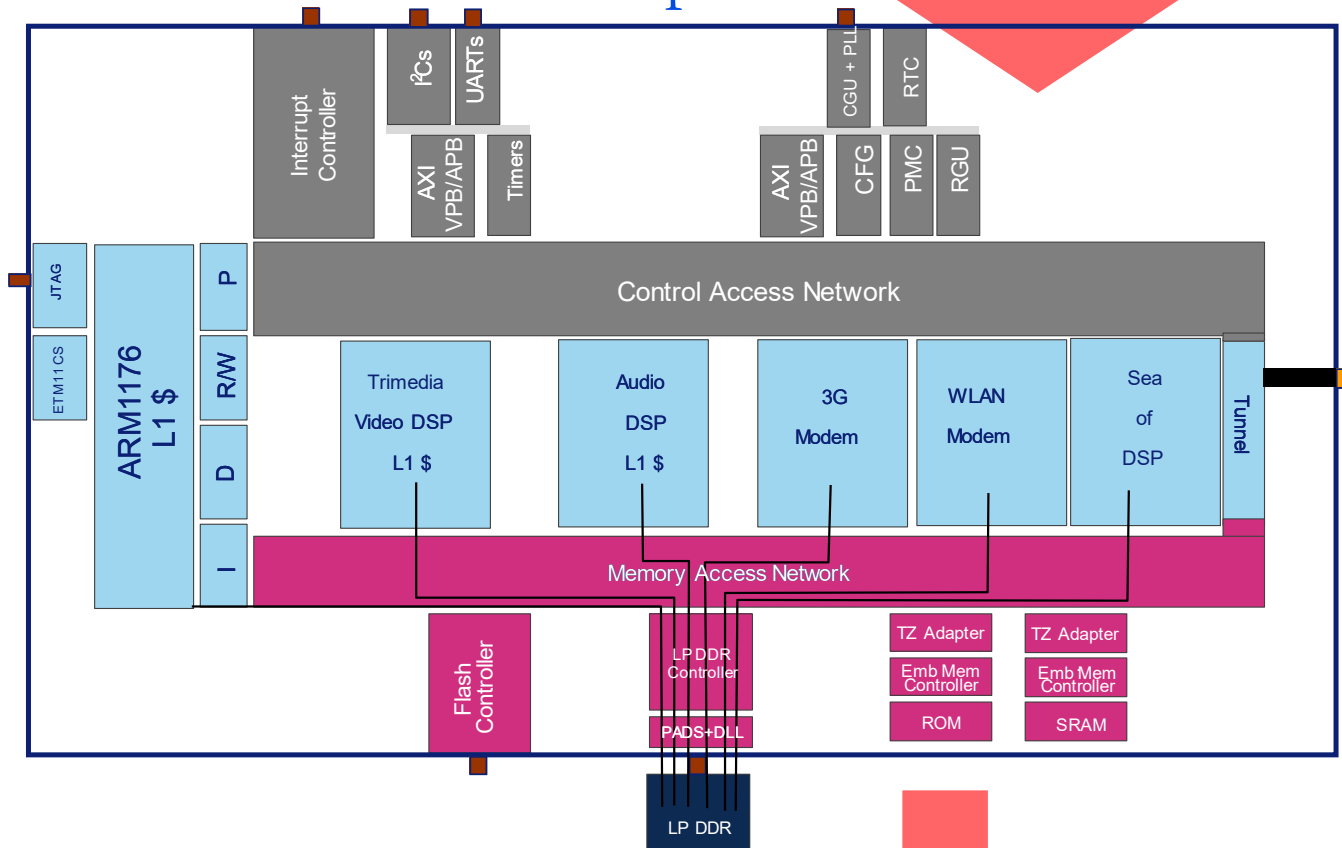
```
1 /* Generated by Yosys 0.7 (git sha1 61f6811, i686-w6
2
3 (* top = 1 *)
4 (* src = "simple.v:1" *)
5 module simple(a, b, c, d);
6   wire _00_;
7   wire _01_;
8   wire _02_;
9   wire _03_;
10  wire _04_;
11  wire _05_;
12  wire _06_;
13  wire _07_;
14  wire _08_;
15  wire _09_;
16  wire _10_;
17  wire _11_;
18  wire _12_;
19  (* src = "simple.v:2" *)
20  input a;
21  (* src = "simple.v:2" *)
22  input b;
23  (* src = "simple.v:2" *)
24  input c;
25  (* src = "simple.v:3" *)
26  output d;
27  NOT _13_ (
28    .A(_05_),
29    .Y(_08_)
30  );
31  NOT _14_ (
32    .A(_06_),
33    .Y(_09_)
34  );
35  NAND _15_ (
36    .A(_09_),
37    .B(_08_),
38    .Y(_10_)
39  );
40  NAND _16_ (
41    .A(_06_),
42    .B(_05_),
43    .Y(_12_)
44  );
45  NAND _17_ (
46    .A(_12_),
47    .B(_10_),
48    .Y(_11_)
49  );
50  assign _05_ = c;
51  assign _06_ = b;
52  assign _07_ = a;
53  assign d = _11_;
54 endmodule
55
```

mycells.lib

```
1 library(demo) {
2   cell(BUF) {
3     area: 6;
4     pin(A) { direction: input; }
5     pin(Y) { direction: output; }
6     function: "A"; }
7
8   cell(NOT) {
9     area: 3;
10    pin(A) { direction: input; }
11    pin(Y) { direction: output; }
12    function: "A'"; }
13
14   cell(NAND) {
15     area: 4;
16     pin(A) { direction: input; }
17     pin(B) { direction: input; }
18     pin(Y) { direction: output; }
19     function: "(A*B)"; }
20
21   cell(NOR) {
22     area: 4;
23     pin(A) { direction: input; }
24     pin(B) { direction: input; }
25     pin(Y) { direction: output; }
26     function: "(A+B)"; }
27
28
29 }
```

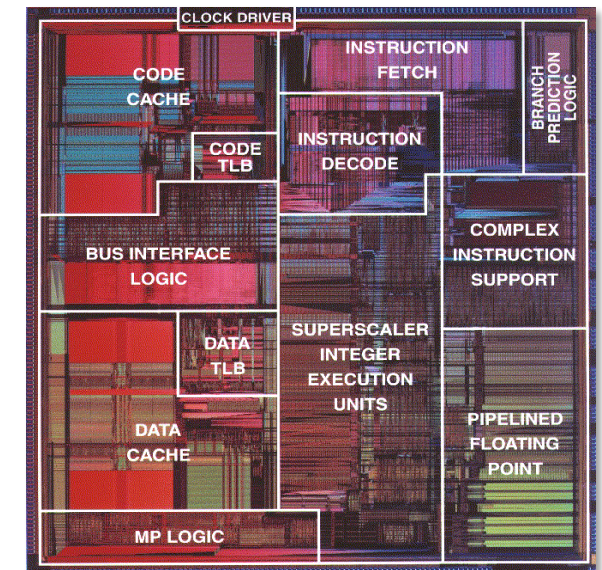


Design Description

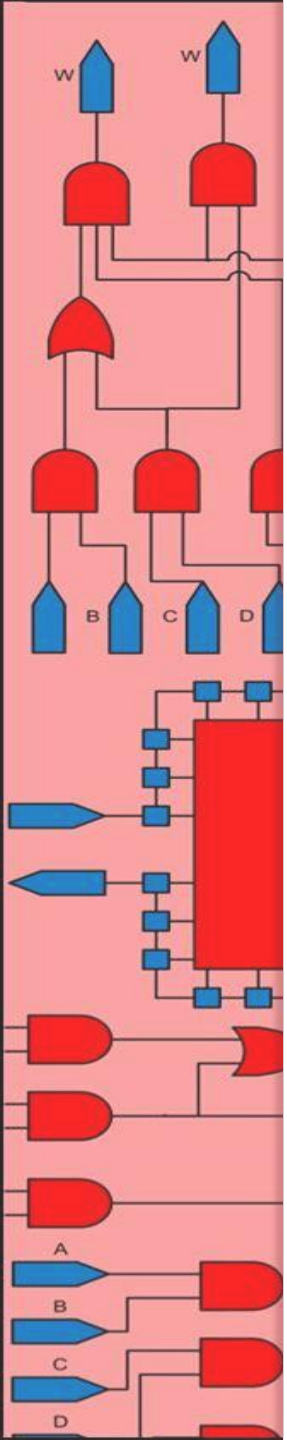


HDL
Model

ASIC/FPGA Implementation



Synthesis



Conclusion

An overview of

- Hierarchical system structure
- SystemVerilog is Hardware description language (HDL)
- Digital design flow